

The Gibbs Phenomenon

Dirichlet Kernel Analysis of Fourier Partial Sums near a Jump Discontinuity

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Abstract

When a piecewise continuous function has a jump discontinuity, its Fourier series does not converge uniformly near that point. The partial sums overshoot and undershoot the function, and this overshoot—the *Gibbs phenomenon*—does not disappear no matter how many Fourier coefficients are used. In this report we first establish the analytic tools needed to study the partial sums: the complex form of the Fourier series, the Dirichlet kernel and its closed form, and the convolution representation $S_N f = f * D_N$. Using these, we give a complete proof of Dirichlet’s theorem on pointwise convergence, including the Riemann–Lebesgue lemma. We then analyze the partial sums of the square wave near its jump and show that, after rescaling, they converge to an integral of the form

$$\frac{2}{\pi} \int_0^u \frac{\sin t}{t} dt,$$

which does not shrink as $N \rightarrow \infty$. As a result, (a) the overshoot approaches a constant that does not depend on N , and (b) this constant equals $(\frac{1}{\pi} \text{Si}(\pi) - \frac{1}{2})\Delta \approx 0.08949 \Delta$, where Δ is the size of the jump. Numerical experiments confirm the limit.

1 Introduction

Fourier series are one of the most fundamental tools in mathematical analysis: they allow a wide class of functions to be expressed as infinite linear combinations of trigonometric functions. For sufficiently smooth periodic functions the Fourier partial sums converge uniformly, and the approximation improves as more terms are included. When the function has a jump discontinuity, however, the convergence behaves quite differently.

A classical observation is that near a jump the Fourier partial sums exhibit a persistent overshoot and undershoot, commonly called the *Gibbs phenomenon* [1, 2]. Although the oscillations become more localized as the number of terms N increases, the height of the overshoot does not decrease; instead it approaches a fixed fraction of the jump size. Explaining why this overshoot persists has been a topic in Fourier analysis for more than a century.

The key to the analysis is the Dirichlet kernel, which appears naturally in the formula for the Fourier partial sums. By examining the kernel near the discontinuity, the partial sums can be described by an integral involving $\frac{\sin t}{t}$, and this integral determines the limiting behaviour of the oscillations. In this report we carry out this argument carefully. Section 2 fixes notation, Section 3 develops the properties of the Dirichlet kernel, Section 4 proves Dirichlet’s theorem, and Section 5 derives the Gibbs constant and verifies it numerically.

2 Fourier Series on $[-\pi, \pi]$

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be 2π -periodic and integrable on $[-\pi, \pi]$. Its Fourier series is

$$a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)), \quad x \in [-\pi, \pi], \quad (2.1)$$

with Fourier coefficients

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx,$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) \, dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) \, dx \quad (n \geq 1).$$

The N -th partial sum is

$$S_N f(x) = a_0 + \sum_{n=1}^N (a_n \cos(nx) + b_n \sin(nx)).$$

If f is smooth, the partial sums approximate f well. If f has a jump discontinuity, however, the Fourier series need not converge to the function value at that point, and the discrepancy must be understood more carefully.

Motivation for Dirichlet's theorem

At a point of discontinuity x_0 , the Fourier series does not converge to $f(x_0)$ itself. Instead, the partial sums converge to a different value, and the way in which they approach it gives rise to the Gibbs phenomenon. To describe the behaviour near such points precisely, we rely on Dirichlet's theorem, stated here and proved in Section 4.

Theorem 2.1 (Dirichlet). *Let f satisfy the following conditions:*

- (i) f is 2π -periodic, i.e. $f(x + 2\pi) = f(x)$ for all $x \in \mathbb{R}$;
- (ii) f is piecewise continuous on $[-\pi, \pi]$;
- (iii) at every point x , the one-sided limits

$$f(x+) = \lim_{t \rightarrow 0^+} f(x+t), \quad f(x-) = \lim_{t \rightarrow 0^+} f(x-t)$$

exist, and so do the one-sided derivatives

$$\alpha_1 = \lim_{t \rightarrow 0^+} \frac{f(x+t) - f(x+)}{t}, \quad \alpha_2 = \lim_{t \rightarrow 0^+} \frac{f(x-t) - f(x-)}{t}.$$

Then for every $x \in \mathbb{R}$,

$$\lim_{N \rightarrow \infty} S_N f(x) = \frac{f(x+) + f(x-)}{2}.$$

Thus, at a jump discontinuity, the Fourier series converges not to $f(x_0)$ but to the average of the left and right limits.

Remark 2.2 (Why the one-sided conditions are needed). The existence of $f(x+)$ and $f(x-)$ is essential: without these limits the value of the Fourier series at the discontinuity cannot be meaningfully identified. The one-sided derivatives control the behaviour of f close to x , which is exactly what is needed to handle the singularity of the Dirichlet kernel at $t = 0$ in the proof below.

3 The Dirichlet Kernel

To prove Theorem 2.1 we first rewrite the partial sums in a convenient analytic form. We begin with two elementary facts that are used repeatedly.

Lemma 3.1 (Integrals of periodic functions). *Let $h : \mathbb{R} \rightarrow \mathbb{C}$ be 2π -periodic and integrable on bounded intervals. Then for every $a \in \mathbb{R}$,*

$$\int_a^{a+2\pi} h(x) dx = \int_{-\pi}^{\pi} h(x) dx.$$

Proof. Choose $k \in \mathbb{Z}$ with $b := -\pi + 2\pi k \leq a < b + 2\pi$. Splitting at $b + 2\pi$,

$$\int_a^{a+2\pi} h(x) dx = \int_a^{b+2\pi} h(x) dx + \int_{b+2\pi}^{a+2\pi} h(x) dx.$$

In the second integral substitute $x = y + 2\pi$ and use $h(y + 2\pi) = h(y)$:

$$\int_{b+2\pi}^{a+2\pi} h(x) dx = \int_b^a h(y) dy.$$

Adding the two pieces gives $\int_b^{b+2\pi} h(x) dx$. Finally, substituting $x = y + 2\pi k$ and using periodicity k times,

$$\int_b^{b+2\pi} h(x) dx = \int_{-\pi+2\pi k}^{\pi+2\pi k} h(x) dx = \int_{-\pi}^{\pi} h(y) dy. \quad \square$$

Lemma 3.2 (Jordan's inequality). *For $0 \leq u \leq \frac{\pi}{2}$, $\sin u \geq \frac{2}{\pi}u$. Consequently, $\sin(t/2) \geq t/\pi$ for $0 \leq t \leq \pi$.*

Proof. Let $\psi(u) = \sin u - \frac{2}{\pi}u$ on $[0, \frac{\pi}{2}]$. Then $\psi(0) = \psi(\frac{\pi}{2}) = 0$ and $\psi''(u) = -\sin u \leq 0$, so ψ is concave. A concave function lies above the chord joining its endpoint values; here the chord is identically 0, so $\psi \geq 0$ on $[0, \frac{\pi}{2}]$. For the second statement put $u = t/2 \in [0, \frac{\pi}{2}]$: $\sin(t/2) \geq \frac{2}{\pi} \cdot \frac{t}{2} = \frac{t}{\pi}$. $\square$

Lemma 3.3 (Complex form). *Define $\hat{f}(m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-imx} dx$ for $m \in \mathbb{Z}$. Then*

$$a_0 = \hat{f}(0), \quad a_n = \hat{f}(n) + \hat{f}(-n), \quad b_n = i(\hat{f}(n) - \hat{f}(-n)) \quad (n \geq 1),$$

and consequently, for every $N \geq 0$,

$$S_N f(x) = \sum_{m=-N}^N \hat{f}(m) e^{imx}.$$

Proof. Step 1: the coefficients. For $m = 0$ we have $e^0 = 1$, so $\hat{f}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f = a_0$. For $n \geq 1$, Euler's formula gives $e^{-inx} = \cos(nx) - i \sin(nx)$ and $e^{inx} = \cos(nx) + i \sin(nx)$. Adding and subtracting,

$$e^{-inx} + e^{inx} = 2 \cos(nx), \quad e^{-inx} - e^{inx} = -2i \sin(nx).$$

Multiplying by $f(x)/(2\pi)$ and integrating over $[-\pi, \pi]$,

$$\begin{aligned} \hat{f}(n) + \hat{f}(-n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) 2 \cos(nx) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx = a_n, \\ \hat{f}(n) - \hat{f}(-n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) (-2i) \sin(nx) dx = -i b_n. \end{aligned}$$

Multiplying the last identity by i gives $b_n = i(\hat{f}(n) - \hat{f}(-n))$.

Step 2: the partial sum. Separate the index $m = 0$ and pair each $m = n \geq 1$ with $m = -n$:

$$\sum_{m=-N}^N \hat{f}(m)e^{imx} = \hat{f}(0) + \sum_{n=1}^N (\hat{f}(n)e^{inx} + \hat{f}(-n)e^{-inx}).$$

Expanding $e^{\pm inx} = \cos(nx) \pm i \sin(nx)$,

$$\begin{aligned} \hat{f}(n)e^{inx} + \hat{f}(-n)e^{-inx} &= (\hat{f}(n) + \hat{f}(-n)) \cos(nx) + i(\hat{f}(n) - \hat{f}(-n)) \sin(nx) \\ &= a_n \cos(nx) + b_n \sin(nx) \end{aligned}$$

by Step 1. Summing over n yields $a_0 + \sum_{n=1}^N (a_n \cos(nx) + b_n \sin(nx)) = S_N f(x)$. $\square$

Definition 3.4 (Dirichlet kernel). For $n \geq 0$,

$$D_n(x) := \sum_{k=-n}^n e^{ikx}.$$

Lemma 3.5 (Real form). $D_n(x) = 1 + 2 \sum_{k=1}^n \cos(kx)$. In particular D_n is real-valued, continuous, even and 2π -periodic.

Proof. Split the sum into $k = 0$, $1 \leq k \leq n$ and $-n \leq k \leq -1$, and in the last part replace k by $-k$:

$$D_n(x) = 1 + \sum_{k=1}^n e^{ikx} + \sum_{k=1}^n e^{-ikx} = 1 + \sum_{k=1}^n (e^{ikx} + e^{-ikx}) = 1 + 2 \sum_{k=1}^n \cos(kx),$$

using $e^{ikx} + e^{-ikx} = 2 \cos(kx)$. The right-hand side is a finite sum of real, continuous, even, 2π -periodic functions, so D_n has all four properties. $\square$

Lemma 3.6 (Closed form). For $x \notin 2\pi\mathbb{Z}$,

$$D_n(x) = \frac{e^{i(n+1)x} - e^{-inx}}{e^{ix} - 1}.$$

Proof. Put $r = e^{ix}$; since $x \notin 2\pi\mathbb{Z}$, $r \neq 1$. Factoring out r^{-n} and reindexing $j = k + n$,

$$D_n(x) = \sum_{k=-n}^n r^k = r^{-n} \sum_{j=0}^{2n} r^j.$$

For the finite geometric sum $G = \sum_{j=0}^{2n} r^j$ we have $rG - G = r^{2n+1} - 1$, because all intermediate powers cancel; hence $G = \frac{r^{2n+1} - 1}{r - 1}$. Therefore

$$D_n(x) = \frac{r^{-n}(r^{2n+1} - 1)}{r - 1} = \frac{r^{n+1} - r^{-n}}{r - 1} = \frac{e^{i(n+1)x} - e^{-inx}}{e^{ix} - 1}. \quad \square$$

Lemma 3.7 (Sine representation). For $x \notin 2\pi\mathbb{Z}$,

$$D_n(x) = \frac{\sin((n + \frac{1}{2})x)}{\sin(x/2)},$$

and $D_n(x) = 2n + 1$ for $x \in 2\pi\mathbb{Z}$, which is also the limit of the right-hand side as $x \rightarrow 2\pi m$.

Proof. Starting from Lemma 3.6, factor $e^{ix/2}$ out of the denominator,

$$e^{ix} - 1 = e^{ix/2}(e^{ix/2} - e^{-ix/2}) = e^{ix/2} \cdot 2i \sin(x/2),$$

and out of the numerator, noting $(n+1) - \frac{1}{2} = n + \frac{1}{2}$ and $-n - \frac{1}{2} = -(n + \frac{1}{2})$:

$$e^{i(n+1)x} - e^{-inx} = e^{ix/2} \left(e^{i(n+\frac{1}{2})x} - e^{-i(n+\frac{1}{2})x} \right) = e^{ix/2} \cdot 2i \sin((n + \frac{1}{2})x).$$

Here we used $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$. For $x \notin 2\pi\mathbb{Z}$ we have $\sin(x/2) \neq 0$, and the common factor $e^{ix/2} \cdot 2i$ cancels, giving the formula.

If $x = 2\pi m$, then $e^{ikx} = 1$ for every k , so $D_n(x) = \sum_{k=-n}^n 1 = 2n+1$ directly from the definition. Since D_n is continuous (Lemma 3.5) and agrees with $\frac{\sin((n+\frac{1}{2})x)}{\sin(x/2)}$ off $2\pi\mathbb{Z}$, this quotient tends to $2n+1$ as $x \rightarrow 2\pi m$. $\square$

Lemma 3.8 (Convolution). *Let f be 2π -periodic and integrable on $[-\pi, \pi]$. Then*

$$S_n f(x) = (f * D_n)(x) := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) D_n(t) dt.$$

Proof. By Lemma 3.3 and the definition of $\hat{f}$,

$$S_n f(x) = \sum_{k=-n}^n \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(s) e^{-iks} ds \right) e^{ikx}.$$

The sum is finite, so it may be moved inside the integral:

$$S_n f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(s) \sum_{k=-n}^n e^{ik(x-s)} ds = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(s) D_n(x-s) ds.$$

Now substitute $s = x - t$, so $ds = -dt$ and $s = -\pi, \pi$ correspond to $t = x + \pi, x - \pi$:

$$S_n f(x) = \frac{1}{2\pi} \int_{x-\pi}^{x+\pi} f(x-t) D_n(t) dt.$$

The integrand $t \mapsto f(x-t)D_n(t)$ is 2π -periodic in t , because both f and D_n are. By Lemma 3.1 with $a = x - \pi$, the integral over $[x - \pi, x + \pi]$ equals the integral over $[-\pi, \pi]$. $\square$

Lemma 3.9 (Integral identity).

$$\int_{-\pi}^{\pi} D_n(x) dx = 2\pi \quad \text{and} \quad \int_0^{\pi} D_n(x) dx = \int_{-\pi}^0 D_n(x) dx = \pi.$$

Proof. For $k \geq 1$, $\int_{-\pi}^{\pi} \cos(kx) dx = \left[\frac{\sin(kx)}{k} \right]_{-\pi}^{\pi} = \frac{\sin(k\pi) - \sin(-k\pi)}{k} = 0$. Hence, by Lemma 3.5,

$$\int_{-\pi}^{\pi} D_n(x) dx = \int_{-\pi}^{\pi} 1 dx + 2 \sum_{k=1}^n \int_{-\pi}^{\pi} \cos(kx) dx = 2\pi.$$

Since D_n is even, the substitution $x \mapsto -x$ shows $\int_{-\pi}^0 D_n = \int_0^{\pi} D_n$. The two halves add up to 2π , so each equals π . $\square$

4 Proof of Dirichlet's Theorem

4.1 The Riemann–Lebesgue lemma

Lemma 4.1 (Riemann–Lebesgue). *Let g be piecewise continuous on $[a, b]$ with one-sided limits at each discontinuity. Then*

$$\lim_{\lambda \rightarrow \infty} \int_a^b g(x) \sin(\lambda x) \, dx = 0,$$

and the same holds with $\cos(\lambda x)$ in place of $\sin(\lambda x)$.

Proof. Step 1: reduction to continuous pieces. Let $a = c_0 < c_1 < \cdots < c_r = b$ consist of a , b and the finitely many discontinuities of g . On each $[c_{j-1}, c_j]$ define g_j to be g on the open interval (c_{j-1}, c_j) , extended to the endpoints by the one-sided limits $g(c_{j-1}+)$ and $g(c_j-)$. Then g_j is continuous on the closed bounded interval $[c_{j-1}, c_j]$, hence bounded and uniformly continuous there. Put

$$M = \max_{1 \leq j \leq r} \sup_{[c_{j-1}, c_j]} |g_j| < \infty.$$

Step 2: a fine partition. Let $\varepsilon > 0$. By uniform continuity there is $\eta > 0$ such that, for every j , $|g_j(x) - g_j(y)| < \varepsilon$ whenever $x, y \in [c_{j-1}, c_j]$ and $|x - y| < \eta$. Subdivide each $[c_{j-1}, c_j]$ into subintervals of length less than η ; together these give a partition

$$P = \{a = x_0 < x_1 < \cdots < x_m = b\}$$

that contains all the c_j . Each open subinterval (x_{k-1}, x_k) lies inside some (c_{j-1}, c_j) , where $g = g_j$. Let ξ_k be the midpoint of $[x_{k-1}, x_k]$. Then

$$|g(x) - g(\xi_k)| < \varepsilon \quad \text{and} \quad |g(\xi_k)| \leq M \quad \text{for all } x \in (x_{k-1}, x_k).$$

The number m depends on ε but not on λ . Values of g at the finitely many points x_k do not affect any integral.

Step 3: splitting the integral. Adding and subtracting $g(\xi_k)$ on each subinterval,

$$\int_a^b g(x) \sin(\lambda x) \, dx = \underbrace{\sum_{k=1}^m \int_{x_{k-1}}^{x_k} (g(x) - g(\xi_k)) \sin(\lambda x) \, dx}_{I_\lambda} + \underbrace{\sum_{k=1}^m g(\xi_k) \int_{x_{k-1}}^{x_k} \sin(\lambda x) \, dx}_{II_\lambda}.$$

Step 4: estimating I_λ . Since $|\sin(\lambda x)| \leq 1$ and $|g(x) - g(\xi_k)| < \varepsilon$ on (x_{k-1}, x_k) ,

$$|I_\lambda| \leq \sum_{k=1}^m \int_{x_{k-1}}^{x_k} \varepsilon \, dx = \varepsilon \sum_{k=1}^m (x_k - x_{k-1}) = \varepsilon(b - a) \quad \text{for every } \lambda > 0.$$

Step 5: estimating II_λ . For $\lambda > 0$,

$$\left| \int_{x_{k-1}}^{x_k} \sin(\lambda x) \, dx \right| = \left| \frac{\cos(\lambda x_{k-1}) - \cos(\lambda x_k)}{\lambda} \right| \leq \frac{2}{\lambda}, \quad \text{hence} \quad |II_\lambda| \leq \sum_{k=1}^m M \cdot \frac{2}{\lambda} = \frac{2Mm}{\lambda}.$$

Step 6: conclusion. Choose $\lambda_0 > 0$ with $2Mm/\lambda_0 < \varepsilon$. For every $\lambda \geq \lambda_0$,

$$\left| \int_a^b g(x) \sin(\lambda x) \, dx \right| \leq |I_\lambda| + |II_\lambda| \leq \varepsilon(b - a) + \varepsilon = \varepsilon(b - a + 1).$$

Since $\varepsilon > 0$ was arbitrary, the limit is 0. For $\cos(\lambda x)$ the same argument applies verbatim, using $\left| \int_{x_{k-1}}^{x_k} \cos(\lambda x) \, dx \right| = \left| \frac{\sin(\lambda x_k) - \sin(\lambda x_{k-1})}{\lambda} \right| \leq \frac{2}{\lambda}$. $\square$

4.2 Proof of Theorem 2.1

Proof. Fix $x \in \mathbb{R}$.

Step 1: folding the integral onto $[0, \pi]$. By Lemma 3.8,

$$S_N f(x) = \frac{1}{2\pi} \int_{-\pi}^0 f(x-t) D_N(t) dt + \frac{1}{2\pi} \int_0^\pi f(x-t) D_N(t) dt.$$

In the first integral substitute $t = -s$; since D_N is even (Lemma 3.5),

$$\int_{-\pi}^0 f(x-t) D_N(t) dt = \int_0^\pi f(x+s) D_N(-s) ds = \int_0^\pi f(x+s) D_N(s) ds.$$

Therefore

$$S_N f(x) = \frac{1}{2\pi} \int_0^\pi (f(x+t) + f(x-t)) D_N(t) dt.$$

Step 2: subtracting the expected limit. By Lemma 3.9, $\int_0^\pi D_N(t) dt = \pi$, so the constant $\frac{f(x+) + f(x-)}{2}$ can be written as

$$\frac{f(x+) + f(x-)}{2} = \frac{1}{2\pi} \int_0^\pi (f(x+) + f(x-)) D_N(t) dt.$$

Subtracting this from Step 1,

$$S_N f(x) - \frac{f(x+) + f(x-)}{2} = \frac{1}{2\pi} \int_0^\pi \phi(t) D_N(t) dt, \quad (4.1)$$

where

$$\phi(t) := f(x+t) - f(x+) + f(x-t) - f(x-).$$

It remains to show that the right-hand side of (4.1) tends to 0 as $N \rightarrow \infty$. By Lemma 3.7, for $0 < t \leq \pi$ we have $D_N(t) = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}$. For $\delta \in (0, \pi)$ we split

$$\frac{1}{2\pi} \int_0^\pi \phi(t) D_N(t) dt = I_N^{(0,\delta]} + I_N^{(\delta,\pi]}, \quad I_N^J := \frac{1}{2\pi} \int_J \phi(t) \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)} dt.$$

Step 3: the part near $t = 0$. By hypothesis (iii),

$$\frac{\phi(t)}{t} = \frac{f(x+t) - f(x+)}{t} + \frac{f(x-t) - f(x-)}{t} \rightarrow \alpha_1 + \alpha_2 \quad (t \rightarrow 0^+).$$

Hence there is $\delta_0 \in (0, \pi)$ such that $|\phi(t)/t - (\alpha_1 + \alpha_2)| \leq 1$, and therefore

$$\left| \frac{\phi(t)}{t} \right| \leq K := |\alpha_1| + |\alpha_2| + 1 \quad \text{for } 0 < t \leq \delta_0.$$

By Lemma 3.2, $\sin(t/2) \geq t/\pi > 0$ for $0 < t \leq \pi$. Together with $|\sin((N+\frac{1}{2})t)| \leq 1$ this gives, for $0 < t \leq \delta_0$,

$$\left| \phi(t) \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)} \right| \leq \frac{|\phi(t)|}{t/\pi} = \pi \left| \frac{\phi(t)}{t} \right| \leq \pi K.$$

Consequently, for every $\delta \in (0, \delta_0]$ and every N ,

$$|I_N^{(0,\delta]}| \leq \frac{1}{2\pi} \int_0^\delta \pi K dt = \frac{K}{2} \delta.$$

Step 4: the part away from $t = 0$. Fix $\delta \in (0, \delta_0]$. On $[\delta, \pi]$, $t/2 \in [\delta/2, \pi/2]$, so $\sin(t/2) \geq \sin(\delta/2) > 0$. The function ϕ is piecewise continuous with one-sided limits, because f is. Dividing by the continuous, nonvanishing function $\sin(t/2)$ preserves these properties, so

$$g(t) := \frac{\phi(t)}{\sin(t/2)}, \quad t \in [\delta, \pi],$$

satisfies the hypotheses of Lemma 4.1. Applying that lemma with $\lambda = N + \frac{1}{2} \rightarrow \infty$,

$$I_N^{(\delta, \pi]} = \frac{1}{2\pi} \int_{\delta}^{\pi} g(t) \sin((N + \frac{1}{2})t) dt \rightarrow 0 \quad (N \rightarrow \infty).$$

Step 5: conclusion. For each $\delta \in (0, \delta_0]$, Steps 3 and 4 give

$$\left| \frac{1}{2\pi} \int_0^{\pi} \phi(t) D_N(t) dt \right| \leq \frac{K}{2} \delta + |I_N^{(\delta, \pi]}|, \quad \text{so} \quad \limsup_{N \rightarrow \infty} \left| \frac{1}{2\pi} \int_0^{\pi} \phi(t) D_N(t) dt \right| \leq \frac{K}{2} \delta.$$

The left-hand side does not depend on δ , and $\delta \in (0, \delta_0]$ is arbitrary. Letting $\delta \rightarrow 0^+$ shows that the lim sup is 0, so the limit exists and equals 0. By (4.1),

$$\lim_{N \rightarrow \infty} S_N f(x) = \frac{f(x+) + f(x-)}{2}. \quad \square$$

5 The Gibbs Phenomenon

Dirichlet's theorem describes the limit at each fixed point. It says nothing about how $S_N f$ behaves at points that move toward the jump as N grows, and this is where the overshoot lives. We analyze the model case of the square wave and then explain why the result is universal.

5.1 The square wave

Let f be the 2π -periodic function with

$$f(x) = \begin{cases} -1, & -\pi < x < 0, \\ 1, & 0 < x < \pi, \end{cases}$$

which has a jump of size $\Delta = f(0+) - f(0-) = 2$ at $x = 0$. Since f is odd, $f(x) \cos(nx)$ is odd and $f(x) \sin(nx)$ is even, so $a_n = 0$ for all $n \geq 0$ and

$$b_n = \frac{2}{\pi} \int_0^{\pi} \sin(nx) dx = \frac{2}{\pi} \left[-\frac{\cos(nx)}{n} \right]_0^{\pi} = \frac{2(1 - (-1)^n)}{\pi n} = \begin{cases} \frac{4}{\pi n}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

Hence, for $N = 2M - 1$ ($M \geq 1$),

$$S_N f(x) = \frac{4}{\pi} \sum_{j=1}^M \frac{\sin((2j-1)x)}{2j-1}.$$

Lemma 5.1. *Let $N = 2M - 1$.*

(a) *For $0 < x < \pi$,*

$$S_N f(x) = \frac{2}{\pi} \int_0^x \frac{\sin(2Mt)}{\sin t} dt.$$

(b) *$S_N f$ is strictly increasing on $[0, \frac{\pi}{2M}]$ and strictly decreasing on $[\frac{\pi}{2M}, \frac{\pi}{M}]$. In particular, $x_M := \frac{\pi}{2M}$ is the first local maximum of $S_N f$ to the right of the jump.*

Proof. (a) $S_N f$ is a finite sum of smooth functions, so it may be differentiated term by term:

$$(S_N f)'(x) = \frac{4}{\pi} \sum_{j=1}^M \cos((2j-1)x).$$

By the product formula $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$ with $A = x$, $B = (2j-1)x$,

$$2 \sin x \cos((2j-1)x) = \sin(2jx) + \sin((2-2j)x) = \sin(2jx) - \sin((2j-2)x).$$

Summing over $j = 1, \dots, M$, the sum telescopes:

$$2 \sin x \sum_{j=1}^M \cos((2j-1)x) = \sin(2Mx) - \sin(0) = \sin(2Mx).$$

For $0 < x < \pi$ we have $\sin x \neq 0$, hence

$$(S_N f)'(x) = \frac{2}{\pi} \cdot \frac{\sin(2Mx)}{\sin x}.$$

The quotient $\frac{\sin(2Mt)}{\sin t}$ extends continuously to $t = 0$ with value $2M$, which equals $\frac{\pi}{2}(S_N f)'(0)$, so $(S_N f)'(t) = \frac{2}{\pi} \frac{\sin(2Mt)}{\sin t}$ holds on $[0, x]$. Every term of $S_N f$ vanishes at 0, so $S_N f(0) = 0$, and the fundamental theorem of calculus gives (a).

(b) For $0 < t < \frac{\pi}{2M}$ we have $0 < 2Mt < \pi$ and $0 < t < \frac{\pi}{2}$, so $\sin(2Mt) > 0$ and $\sin t > 0$, hence $(S_N f)'(t) > 0$. For $\frac{\pi}{2M} < t < \frac{\pi}{M}$ we have $\pi < 2Mt < 2\pi$, so $\sin(2Mt) < 0$, while still $0 < t < \pi$ and $\sin t > 0$, hence $(S_N f)'(t) < 0$. (When $M = 1$ the second interval is $(\frac{\pi}{2}, \pi)$, where $\sin t > 0$ as well.) This proves (b). $\square$

Note that $x_M \rightarrow 0$: the peak moves toward the discontinuity as N grows.

Theorem 5.2 (Gibbs constant). *Let $\text{Si}(u) = \int_0^u \frac{\sin t}{t} dt$. For each fixed $u \in (0, \pi]$,*

$$\lim_{M \rightarrow \infty} S_{2M-1} f\left(\frac{u}{2M}\right) = \frac{2}{\pi} \text{Si}(u).$$

In particular, with $x_M = \frac{\pi}{2M}$,

$$\lim_{M \rightarrow \infty} S_{2M-1} f(x_M) = \frac{2}{\pi} \text{Si}(\pi) \approx 1.178980.$$

Proof. Fix $u \in (0, \pi]$ and $M \geq 1$. Then $x = \frac{u}{2M} \in (0, \frac{\pi}{2}] \subset (0, \pi)$, so Lemma 5.1(a) applies. Substituting $t = \frac{s}{2M}$, $dt = \frac{ds}{2M}$,

$$S_{2M-1} f\left(\frac{u}{2M}\right) = \frac{2}{\pi} \int_0^u \frac{\sin s}{\sin(s/(2M))} \cdot \frac{ds}{2M} = \frac{2}{\pi} \int_0^u h_M(s) ds, \quad h_M(s) := \frac{\sin s}{2M \sin(s/(2M))}.$$

Pointwise limit. For fixed $s \in (0, u]$, write

$$2M \sin\left(\frac{s}{2M}\right) = s \cdot \frac{\sin(s/(2M))}{s/(2M)} \rightarrow s \cdot 1 = s \quad (M \rightarrow \infty),$$

because $\frac{\sin y}{y} \rightarrow 1$ as $y \rightarrow 0$. Hence $h_M(s) \rightarrow \frac{\sin s}{s}$.

Uniform bound. For $s \in (0, u]$ we have $\frac{s}{2M} \in (0, \frac{\pi}{2}]$, so Lemma 3.2 gives $\sin(\frac{s}{2M}) \geq \frac{2}{\pi} \cdot \frac{s}{2M}$, i.e. $2M \sin(\frac{s}{2M}) \geq \frac{2s}{\pi}$. Using $|\sin s| \leq s$ for $s \geq 0$,

$$|h_M(s)| \leq \frac{|\sin s|}{2s/\pi} = \frac{\pi}{2} \cdot \frac{|\sin s|}{s} \leq \frac{\pi}{2} \quad \text{for all } s \in (0, u], \quad M \geq 1.$$

Passing to the limit. The functions h_M are continuous on $(0, u]$, uniformly bounded by the constant $\frac{\pi}{2}$ (which is integrable on the bounded interval $(0, u]$), and converge pointwise to $\frac{\sin s}{s}$. By the dominated convergence theorem (or Arzelà's bounded convergence theorem for Riemann integrals),

$$\lim_{M \rightarrow \infty} \int_0^u h_M(s) \, ds = \int_0^u \frac{\sin s}{s} \, ds = \text{Si}(u).$$

Multiplying by $\frac{2}{\pi}$ proves the first claim. Taking $u = \pi$ and using $\text{Si}(\pi) \approx 1.8519370$ gives $\frac{2}{\pi} \text{Si}(\pi) \approx 1.178980$. $\square$

Corollary 5.3 (The overshoot does not vanish). *(a) The overshoot of the square-wave partial sums at their first peak satisfies*

$$\lim_{M \rightarrow \infty} (S_{2M-1}f(x_M) - f(0+)) = \frac{2}{\pi} \text{Si}(\pi) - 1 = \left(\frac{1}{\pi} \text{Si}(\pi) - \frac{1}{2} \right) \Delta \approx 0.08949 \Delta.$$

(b) The undershoot to the left of the jump has the same size:

$$\lim_{M \rightarrow \infty} (f(0-) - S_{2M-1}f(-x_M)) = \frac{2}{\pi} \text{Si}(\pi) - 1.$$

(c) The convergence $S_N f \rightarrow f$ is not uniform on $(0, \pi)$: $\liminf_{M \rightarrow \infty} \sup_{0 < x < \pi} |S_{2M-1}f(x) - f(x)| \geq \frac{2}{\pi} \text{Si}(\pi) - 1 > 0$.

Proof. (a) Since $f(0+) = 1$, Theorem 5.2 gives $S_{2M-1}f(x_M) - 1 \rightarrow \frac{2}{\pi} \text{Si}(\pi) - 1$. With $\Delta = 2$,

$$\left(\frac{1}{\pi} \text{Si}(\pi) - \frac{1}{2} \right) \Delta = \frac{2}{\pi} \text{Si}(\pi) - 1 \approx 1.178980 - 1 = 0.178980 = 0.08949 \times 2.$$

(b) $S_{2M-1}f$ is a sum of sines, hence odd, so $S_{2M-1}f(-x_M) = -S_{2M-1}f(x_M)$. Since $f(0-) = -1$,

$$f(0-) - S_{2M-1}f(-x_M) = -1 + S_{2M-1}f(x_M) \rightarrow \frac{2}{\pi} \text{Si}(\pi) - 1.$$

(c) Since $x_M \in (0, \pi)$ and $f(x_M) = 1$,

$$\sup_{0 < x < \pi} |S_{2M-1}f(x) - f(x)| \geq S_{2M-1}f(x_M) - 1,$$

and the right-hand side tends to $\frac{2}{\pi} \text{Si}(\pi) - 1 \approx 0.17898 > 0$ by (a). A uniformly convergent sequence would have this supremum tend to 0. $\square$

So the relative overshoot is about 8.949% of the jump, independent of N . The convergence $S_N f(x) \rightarrow f(x)$ at each fixed $x \neq 0$ (Theorem 2.1) is therefore not uniform near the jump (Corollary 5.3(c)): the peak of height ≈ 1.179 simply moves closer to $x = 0$ instead of flattening out.

5.2 Numerical verification

Table 1 lists the maximum of $S_N f$ on $(0, \pi)$ computed numerically. In every case the maximum is attained at $x_M = \pi/(2M)$, and the relative overshoot converges to 0.08949. Figure 1 shows the partial sums near the jump and the convergence of the relative overshoot.

N	$\max S_N f - 1$	overshoot/ Δ	location/ x_M
1	0.273240	0.136620	1.000
5	0.188357	0.094178	1.000
15	0.180284	0.090142	1.000
39	0.179188	0.089594	1.000
99	0.179013	0.089507	1.000
299	0.178983	0.089492	1.000
999	0.178980	0.089490	1.000
∞	$\frac{2}{\pi} \text{Si}(\pi) - 1 = 0.178980$	0.089490	

Table 1: Overshoot of the square-wave partial sums ($\Delta = 2$).

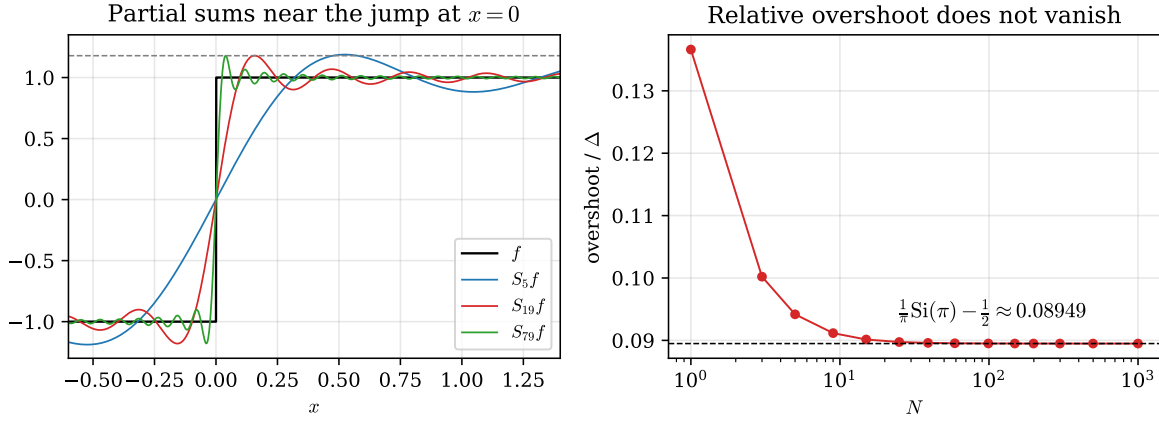


Figure 1: Left: partial sums $S_N f$ of the square wave for $N = 5, 19, 79$; the dashed line marks the limiting peak height $\frac{2}{\pi} \text{Si}(\pi) \approx 1.179$. Right: relative overshoot versus N (log scale), converging to 0.08949.

5.3 Universality

Remark 5.4. The constant is the same for every function satisfying the hypotheses of Theorem 2.1. If f has a jump of size $\Delta = f(x_0+) - f(x_0-)$ at x_0 , let h denote the square wave above and set

$$g(x) = f(x) - \frac{\Delta}{2} h(x - x_0).$$

Then g has no jump at x_0 , and for piecewise smooth g it is a standard result that its Fourier series converges uniformly on a neighbourhood of x_0 (see e.g. [3]). Since $S_N f = S_N g + \frac{\Delta}{2} S_N h(\cdot - x_0)$, the overshoot of $S_N f$ near x_0 is asymptotically that of $\frac{\Delta}{2} S_N h$, namely $\frac{\Delta}{2} (\frac{2}{\pi} \text{Si}(\pi) - 1) \approx 0.08949 \Delta$. We do not reprove the uniform convergence result here.

6 Conclusion

Rewriting the Fourier partial sums as a convolution with the Dirichlet kernel reduces both questions—pointwise convergence and the behaviour near a jump—to the analysis of the integral

$$\int_0^\pi \phi(t) \frac{\sin((N + \frac{1}{2})t)}{\sin(t/2)} dt.$$

Away from $t = 0$, the Riemann–Lebesgue lemma makes the oscillating kernel average out, which yields Dirichlet’s theorem. Near $t = 0$, however, the kernel behaves like $\frac{\sin(Nt)}{t}$, and after rescaling t by N the partial sums converge to $\frac{2}{\pi} \text{Si}(u)$, a profile that does not depend on N . Its

first maximum, $\frac{2}{\pi} \text{Si}(\pi)$, gives an overshoot of $(\frac{1}{\pi} \text{Si}(\pi) - \frac{1}{2})\Delta \approx 0.08949 \Delta$. This explains both why the Gibbs phenomenon persists and why the overshoot ratio is universal for Fourier series.

References

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